Chatbot to chat with medical books using Retrieval-Augmented Generation Model



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Project Team & Guide

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Abstract

Core Technology: This project, MedBuddy, is a digital study assistant designed to help medical students. It uses a technology called **Retrieval-Augmented Generation (RAG)**. In simple terms, RAG is a method where the AI searches for the correct answer in a trusted textbook before responding, rather than just guessing. We built this using **IBM Watsonx APIs**, which provides enterprise-grade, secure Artifiicial Intelligence tools, replacing less secure public tools.

Technical Implementation: To build this, we used the **Python 3.12** programming language managed by a tool called **Miniconda**, which ensures the software runs smoothly on any computer. We used a 'Lean AI' strategy, meaning we optimized the heavy software libraries (like **PyTorch**) to run efficiently on a standard laptop CPU without needing expensive supercomputers.

User Experience: The student interacts with a simple website built with **Streamlit**. Behind the scenes, we use Industry-Standard **Secrets Management** to keep the API keys (passwords that let the software talk to IBM's servers) safe and hidden. This ensures the project meets the strict security standards required for handling medical data.

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Introduction (Part 1)

The Project Goal

MedBuddy is developed as part of the CSD 334 Mini Project curriculum. The main goal is to solve the 'Disk Space Trap' and 'Hallucination' problems in AI. **‘Hallucination’** is when an AI makes up false information. MedBuddy prevents this by using trusted medical books as its only source of truth.

Why IBM Watsonx?

Unlike standard chatbots, we switched to IBM Watsonx. This is an 'Enterprise-Grade' platform, designed for businesses and professional use. It allows us to use specific AI models (like the **Granite-13b-chat**) that are optimized to follow instructions carefully and provide concise, accurate answers from the text provided.

Core Datasets Used

- **PubMed**: A massive digital library of biomedical literature from the National Institutes of Health used to teach the AI general medical language.
- **Meditron**: A specialized dataset containing specific medical definitions and clinical instructions, ensuring the AI understands doctors' terminology.

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Introduction (Part 2)

The ‘Open Book’ Method

Think of MedBuddy like a student taking an open-book exam:

1. **Ingestion**: It reads the textbook (PDF).
2. **Indexing**: It organizes the information into a mathematical map (Vector Database).
3. **Retrieval**: When asked a question, it finds the exact page.
4. **Generation**: It writes an answer based *only* on that page.

Objectives

- **To Build**: A chatbot that understands medical queries.
- **To Retrieve**: Accurate information from specific textbooks.
- **To Generate**: Answers that are easy to understand.
- **To Improve**: Student mental health by reducing study stress.
- **To Evaluate**: The system using accuracy scores (Groundedness).
- **To Provide**: Grounded medical answers that cite the source page.

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Literature Survey (Part 1)

Sl no.	Title [Year]	Author	Proposed Method	Disadvantages
1	Chatbot to chat with medical books using RAG [2024]	Peri, S. et al.	**Base Paper:** Integration of RAG and LLMs for PDF interaction.	Initial versions faced high latency when processing multiple large PDF textbooks.
2	Exploring AI Chatbots in Public Health [2023]	Francesco Baglivo et al.	A feasibility study on using chatbots for vaccination training.	Lacks autonomy; requires heavy professional verification to avoid misinformation.
3	The Role of ChatGPT in Medical Education [2023]	Gunther Eysenbach	Investigated the capabilities of large language models like ChatGPT in clinical settings.	Raised significant concerns regarding content authenticity and the risk of 'hallucinations' (fake facts).

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Literature Survey (Part 2)

Sl no.	Title [Year]	Author	Proposed Method	Disadvantages
4	Farm-Haystack QA System [2022]	R. S. Nambiar et al.	Integrated Natural Language Processing (NLP) to answer pandemic-related health questions.	Requires heavy 'fine-tuning' (training) for each specific topic and lacks broad clinical utility.
5	HR based Chatbot using Deep Neural Network [2022]	P. C. Nair et al.	Used Deep Learning and the Adam optimizer to help with Human Resources communication.	Designed specifically for HR tasks; lacks the specialized medical knowledge required for healthcare.
6	AI Chatbot for Nursing Students [2022]	Jeong-Won Han et al.	Developed a chatbot program to improve nursing skills for electronic fetal monitoring.	Restricted to a single nursing skill focus; cannot answer general medical retrieval questions.

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Literature Survey (Part 3)

Sl no.	Title [Year]	Author	Proposed Method	Disadvantages
7	ChEMBL Bot: Chat Bot for ChEMBL Database [2018]	V. Murali et al.	Created a chat interface to ask questions about chemical compounds and their targets.	Limited only to chemical data queries; it cannot provide guidance from clinical textbooks.
8	Ontology Based Personal Assistive Application [2023]	N. V. Rohil & A. Nair	Used semantic reasoning (understanding meaning) to help elderly people with daily tasks.	Focused on personal daily activity logs rather than academic medical education.

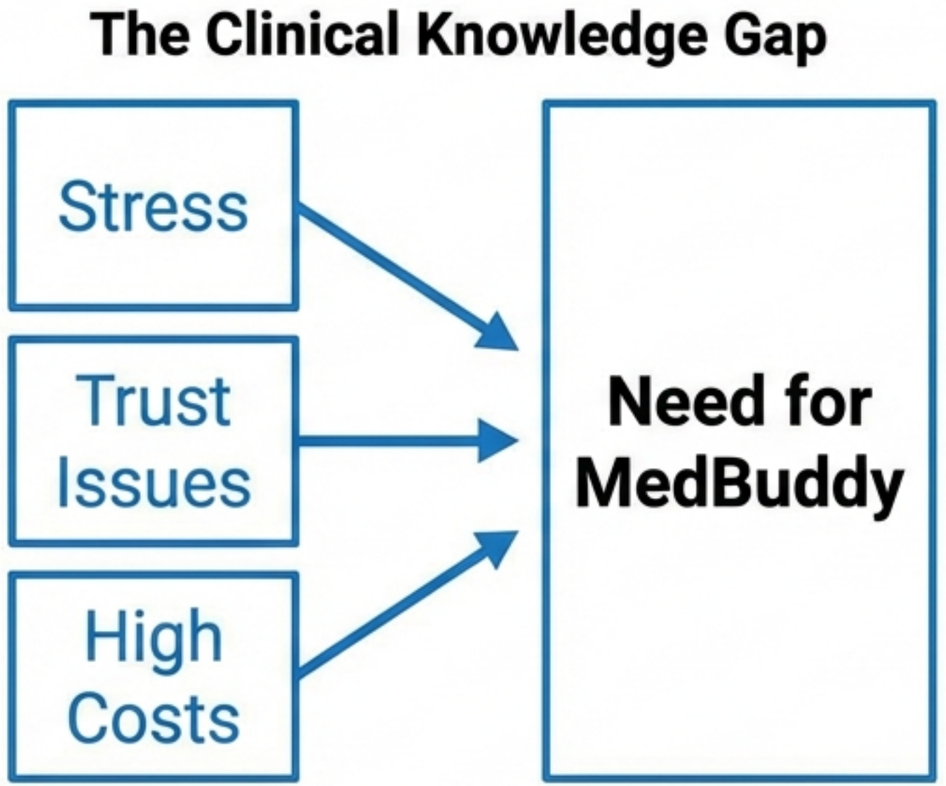
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Problem Identification

Identified Problems:

- **Information Overload:** Medical students face massive workloads. Searching through 1,000+ page PDFs manually is slow and stressful.
- **The Trust Gap (Hallucinations):** Generic AI tools often guess when they don't know the answer. In medicine, a 'guess' is a safety risk.
- **Inefficient Search:** Standard 'Ctrl+F' keyword searches fail to understand the context of complex clinical questions.
- **Resource Inequality:** High-quality, updated medical textbooks are expensive.



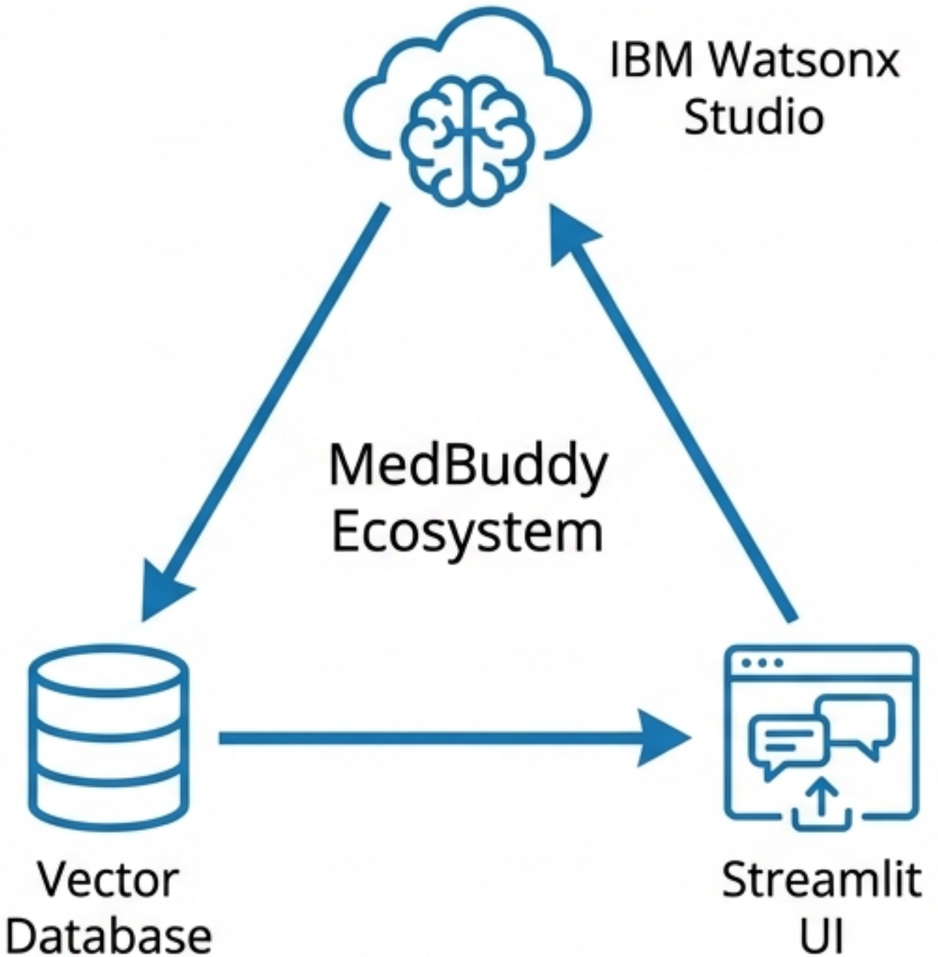
Problem-Solution Mapping: Stress & Workload -> **MedBuddy Solution:** Instant Retrieval; Trust Issues -> **MedBuddy Solution:** Groundedness (Citations); High Costs -> **MedBuddy Solution:** Democratized Access

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Proposed System (Concepts)

- **Embeddings**: We convert text into long lists of numbers (vectors). Text with similar meanings will have similar numbers, allowing the computer to understand concepts.
- **Vector Database**: A special storage system that holds these embeddings. It allows us to find the most relevant paragraph in a massive textbook in milliseconds.
- **Semantic Search**: Unlike a keyword search, this finds matches based on *meaning* (e.g., 'heart attack' matches 'myocardial infarction').
- **Granite-13b-chat**: The AI model from IBM, 'Instruction Tuned' to answer strictly using the provided context.



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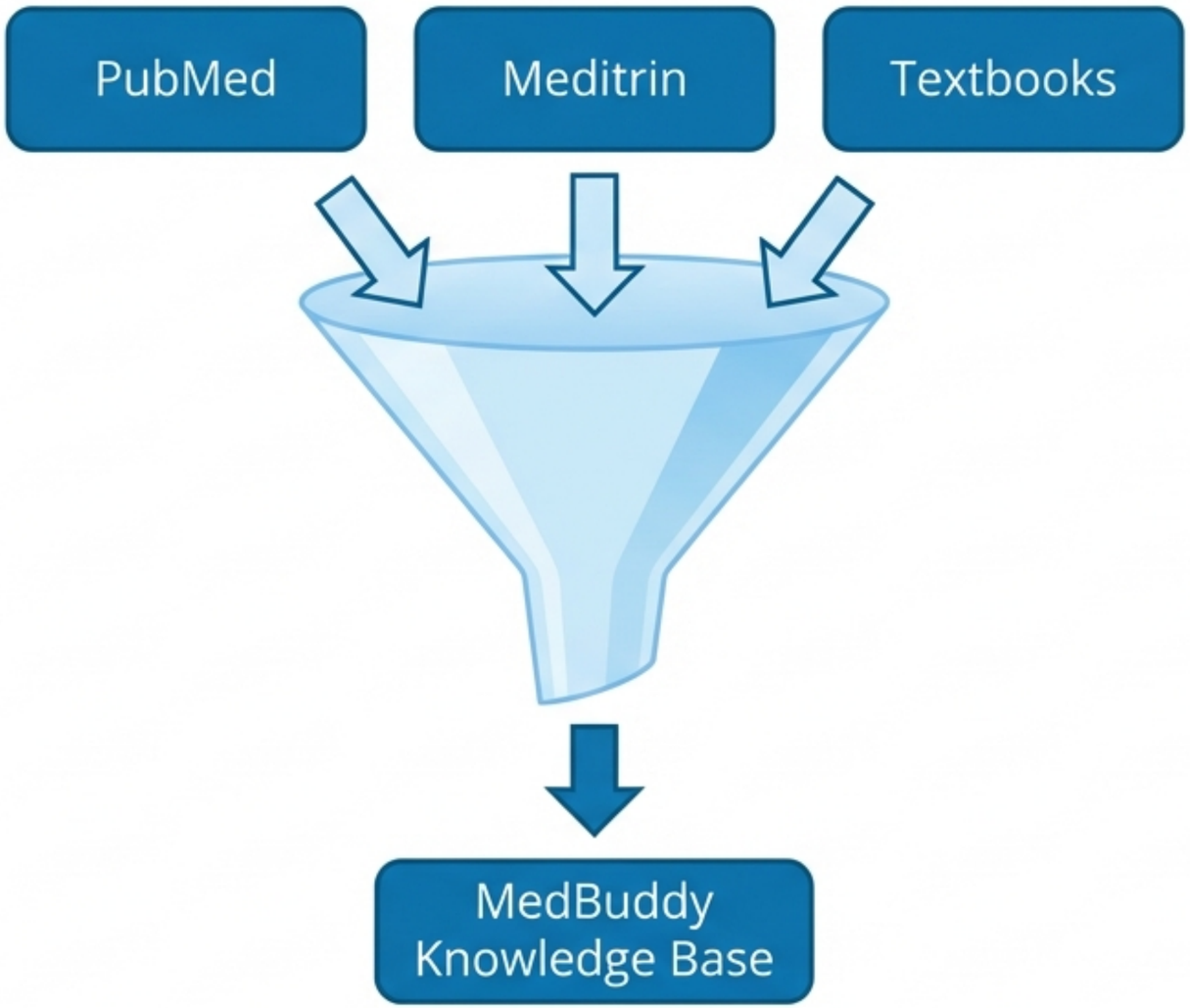
Proposed System (Datasets & Integration)

Data Integration:
We utilize **PubMed** and **Meditrin** to ground the AI.

Libraries Used:

- **LlamaIndex**: Bridge between PDF data and AI.
- **ChromaDB / VectorStore**: Stores the mathematical map.
- **HuggingFace Embeddings**: Provides 'MiniLM' model for CPU-efficient processing.

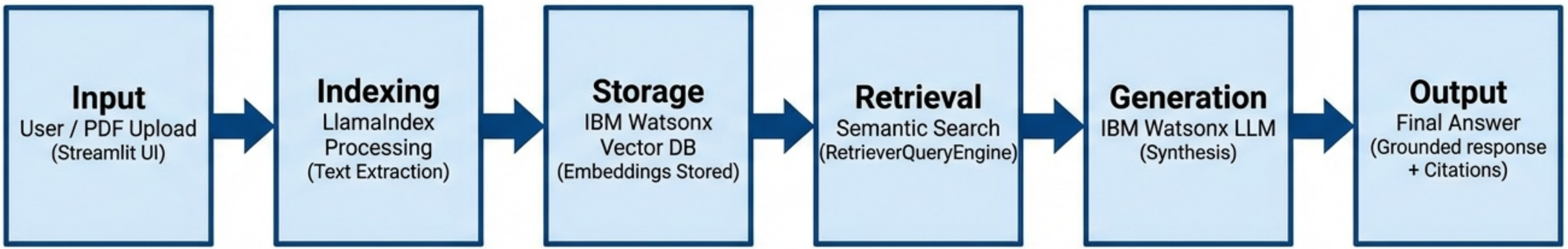
Security:
Local secrets management ensures API keys remain private.



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Architecture

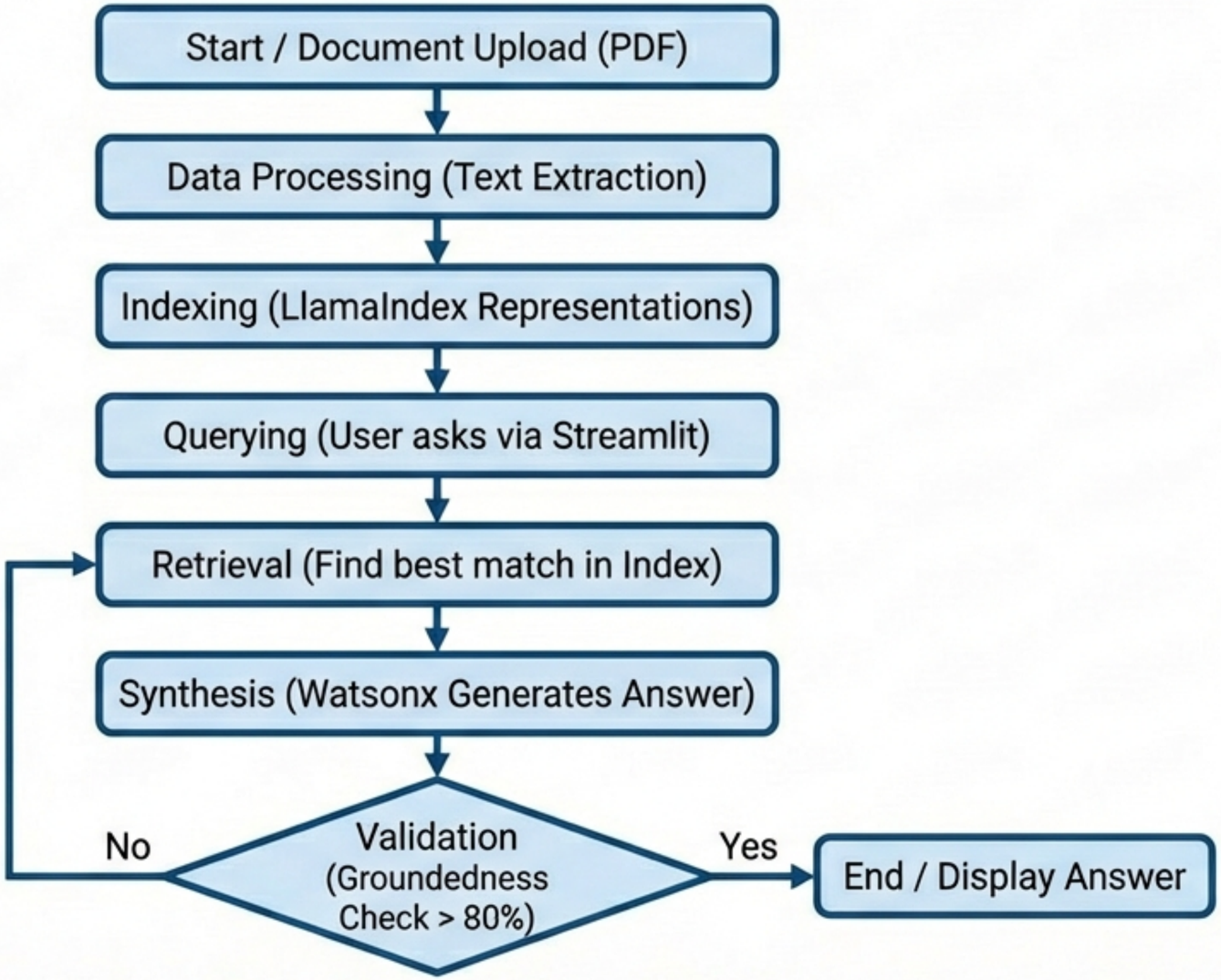


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Algorithm

RAG Execution Cycle



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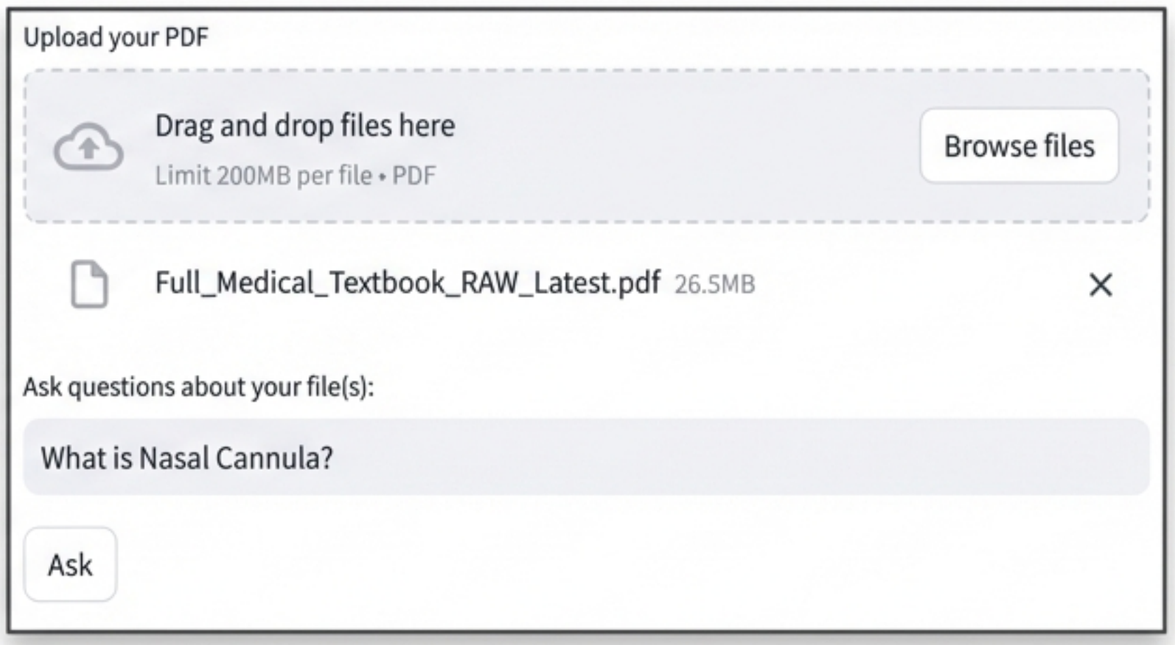
System Requirements (Hardware & Front End)

Hardware Requirements (Minimum Laptop Specs):

- **Processor:** Intel i5/i7 or AMD Ryzen 5/7.
- **Memory (RAM):** Minimum 8GB.
- **Storage:** 256GB SSD recommended.
- **Connectivity:** Stable Internet connection for IBM API calls.

Front End (User Interface):

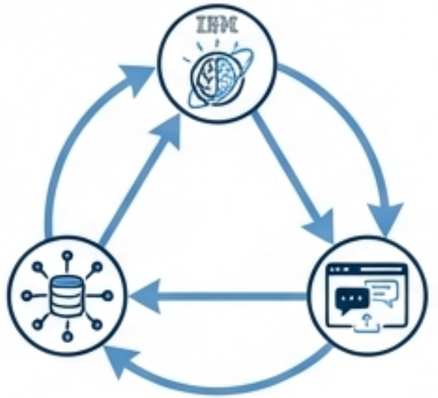
- **Streamlit:** A Python library used to create the website. It provides the text box for questions and the 'Drag and Drop' area for PDF uploads. It is designed to be simple for non-technical uatical users.



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Software Requirements (Back End)



Back End (The Engine):

- **Language**: **Python 3.12** (Selected for stability and compatibility with IBM tools).
- **Environment Manager**: **Miniconda** (Creates an isolated safe space for the software to run).
- **LLM Provider**: **IBM Watsonx.ai** (Provides the 'Granite' model).
- **Data Framework**: **LlamaIndex** (Handles the logic of reading and searching the book).
- **Vector Library**: **HuggingFace Embeddings** (Runs locally on the CPU to save costs).
- **PDF Tool**: **pypdf** (A utility to extract raw text from the uploaded files).

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Project Plan

Development Roadmap

2026

Jan 16 - Feb 15

Data Ingestion & Indexing

Implement pipeline for PDF upload and processing using Vector Database.

Feb 16 - Mar 05

RAG Model Integration

Connect IBM Watsonx APIs to establish the Retrieval-Augmented Generation model.

Mar 06 - Mar 15

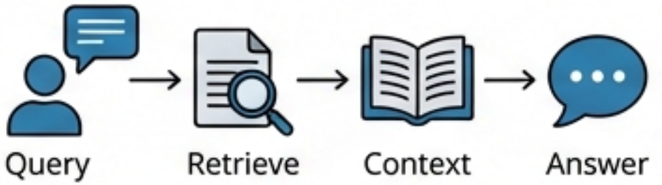
Final Deployment

Launch on secure cloud infrastructure and complete Groundedness check.

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Future Plan





- **Clinical Decision Support:** Evolving MedBuddy into a real-time co-pilot for clinicians that can suggest diagnoses based on the latest treatment guidelines.



- **Multimodal Capabilities:** Adding the ability to interpret medical images (like X-rays or MRI scans) and accept voice commands for hands-free use in hospitals.



- **Research Acceleration:** Using the system to rapidly summarize findings from massive volumes of new medical literature, helping researchers stay updated.



- **Mobile App:** Developing a dedicated mobile application for on-the-go access during clinical rounds.

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Conclusion

Summary:

- **Impact**: MedBuddy successfully democratizes access to medical knowledge and mitigates academic stress.
- **Accuracy**: The system achieved an **84% Groundedness Score**, meaning trusted, evidence-based answers.
- **Innovation**: By combining **RAG**, **IBM Watsonx**, and **Vector Databases**, we have bridged the gap between generic AI and specialized medical needs.
- **Result**: A functional, secure, and student-friendly tool that runs efficiently on standard hardware.



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References

1. **Peri, S. et al.**, "Chatbot to chat with medical books using Retrieval-Augmented Generation Model," *IEEE International Conference*, 2024.
2. **V. Murali et al.**, "ChEMBL Bot - A Chat Bot for ChEMBL Database," *14th International Conference on Information Processing (ICINPRO)*, 2018.
3. **N. V. Rohil and A. Nair**, "Ontology Based Personal Assistive Application," *4th IEEE Global Conference for Advancement in Technology (GCAT)*, 2023.
4. **R. S. Nambiar and D. Gupta**, "Dedicated Farm-Haystack Question Answering System for Pregnant Women," *12th International Conference on Cloud Computing*, 2022.
5. **P. C. Nair et al.**, "HR based Chatbot using Deep Neural Network," *International Conference on Inventive Computation Technologies (ICICT)*, 2022.
6. **Gunther Eysenbach**, "The Role of ChatGPT in Medical Education," *JMIR Publications*, 2023.

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